
Weight-Dictionary Decomposition: Reading a Transformer’s Residual Stream with Its Own Writes

Abstract

1 Sparse autoencoders interpret transformer residual states with an overcomplete
2 dictionary trained on activations. We build the dictionary from the weights instead.
3 Each component’s contribution to the residual stream is confined to a direction or
4 subspace fixed by its weights, so collecting these directions from the checkpoint
5 yields an overcomplete dictionary, fitted to no data, whose atoms carry deterministic
6 provenance to a component or its write subspace. Weight-dictionary decomposition
7 (WDD) sparse-codes an observed state over this dictionary to infer which native
8 writes are most consistent with it. Scored on ground-truth forward passes in five
9 model families, coefficient estimates for correctly identified writes correlate almost
10 perfectly with the truth ($r = 0.998$; 0.648 when misses count as zero), while
11 identification is capped by an observability ceiling: once later computation cancels
12 a write along its own direction, projection from the later state cannot recover its
13 coefficient. The construction blindly recovers both of OLMo-1B’s published super
14 weights. Used as measurement axes, the write directions single out a GPT-2 neuron
15 that seems to erase the model’s massive activation channel, and interventions
16 reinterpret it as a participant in its maintenance, a reading that survives frozen
17 normalization. In pre-registered comparisons WDD’s hit rate merely matches
18 a correlation screen, and its value is the testable mechanistic content of each
19 nomination.

20 1 Introduction

21 Interpreting a transformer’s residual stream currently runs through trained sparse dictionaries: sparse
22 autoencoders (SAEs) learn an overcomplete basis, whose elements are conventionally called *atoms*,
23 from millions of activation samples [3, 5]. Training is the expense, and the learned atoms are the
24 interpretive risk: a direction that reconstructs activations well need not correspond to anything the
25 model does.

26 A structural fact suggests an alternative. As the circuits literature established [7], the residual stream
27 is a sum of writes through fixed weight matrices, so each component’s write is confined in advance
28 to a direction or subspace its weights determine: an MLP neuron along one fixed direction with an
29 input-dependent coefficient, an attention head inside its output-projection row space, the embedding
30 along fixed vectors. Prior work interprets these weight vectors directly [8, 6, 12] or tracks a known
31 write forward through the stream [10]. To our knowledge, no prior method inverts the relationship:
32 given one observed state and the checkpoint, infer which native writes are most consistent with the
33 state, and at what strength.

34 We evaluate this inversion as an instrument, scored on exact ground truth from hooked forward passes,
35 and as a discovery tool whose hypotheses we test by intervention. Our contributions:

- 36 1. **A training-free, provenance-bearing dictionary, validated as an instrument.** Reconstruction
37 is strong with no training, and a rotation control traces this to coordinate alignment. Attribution
38 splits into near-exact estimation and identification capped by a measurable observability ceiling.

Both replicate across five families and two corpora, with estimation accurate in every decile for gated-MLP models.

2. **A discovery pipeline with intervention discipline.** A null-tested write-role taxonomy nominates a GPT-2 neuron whose apparent erasure of the massive activation channel resolves, under intervention, into participation in maintaining that channel, with a suppression field holding back a reserve bank.
3. **Pre-registered nomination comparisons, reported in full.** WDD does not outperform a one-line correlation baseline on causal-coupling hit rate. Its proposed advantage is representational: each nomination carries a native write direction, a predicted partner, and a falsifiable role hypothesis.

2 Method

Dictionary and inverse problem. An MLP neuron with down-projection row $\mathbf{w}_j$ and post-nonlinearity activation a_j writes $a_j \mathbf{w}_j = c_j \mathbf{d}_j$ into the stream, with unit direction $\mathbf{d}_j = \mathbf{w}_j / \|\mathbf{w}_j\|$ and coefficient $c_j = a_j \|\mathbf{w}_j\|$. We collect unit-normed atoms for all model-native directions whose writes can reach the target state: token and position embedding rows (contributed before block 0), an SVD basis of each attention head’s attainable output subspace, every MLP down-projection row, and per-block biases. Each atom maps deterministically to its (block, component, write subspace). MLP atoms correspond one-to-one with neuron write directions, while attention atoms form a basis of a head’s attainable subspace and are not individual causal writes. For GPT-2 small at layer 6 (blocks 0–6) this yields $m = 78,175$ atoms (approximately $102\times$ overcomplete). WDD solves the inverse sparse mixing problem: given a state $\mathbf{x}$ and $D = [\mathbf{d}_1 \cdots \mathbf{d}_m]$, estimate $\hat{\mathbf{c}} = \text{OMP}_k(D, \mathbf{x})$ with $\mathbf{x} \approx D\hat{\mathbf{c}}$, reading each $\hat{c}_i$ as the inferred write of atom i ’s component. All control dictionaries share the identical OMP code and size. GPT-2 stores projections in Conv1D form, whose rows are the write directions; where the down-projection is a standard linear layer, its columns play that role, and gating changes only the coefficients. Identification mode ranks atoms by one-shot correlation and refits on that support (Sec. 4.2).

Survival and write roles. For a write with unit direction $\mathbf{d}$ and signed coefficient c , survival in a later state $\mathbf{x}$ is $s = (\mathbf{x} \cdot \mathbf{d})/c$. This is the projection ratio of Janiak et al. [10]. Birth survival obeys $s_{\text{birth}} = 1 + (\mathbf{x}_{\text{pre}} \cdot \mathbf{d})/c$ and classifies roles: born $s > 1$, *reinforcing*; $s \approx 1$, *novel*; $s < 0$, *counter-aligned*, opposing existing content along its own direction. The labels record what a component wrote relative to the state. Assigning mechanism requires intervention.

3 Experimental Setup

Our primary model is GPT-2 small. We decompose the post-block-6 state over 32,768 held-out WikiText-2 tokens (clean cross-entropy 3.647). Train-split tokens supply centering means and data-based controls. We rebuild the dictionary in SmolLM2-135M, Qwen2.5-0.5B, Pythia-410m, and OLMo-1B at their middle layers, and replicate the core evaluation on a Pile-derived corpus. Ground truth is the *ledger*: every MLP neuron’s true write coefficient (post-nonlinearity activation times row norm) from hooked forward passes. Attention writes carry no per-component scalar, so attribution is scored on MLP writes. A token’s *dominant writes* are its top-3 ledger entries by magnitude.

4 Results

4.1 Reconstruction Is Not Attribution

Figure 1 (left) sweeps sparsity. The rotation control shows the advantage comes from alignment with the model’s residual coordinate system. The critical negative result is a provenance-null control. A dictionary whose atoms could not have contributed, drawn only from future blocks 7–11, reconstructs nearly as well as a size-matched correct one (FVU 0.091 vs. 0.073 at $k=32$), and ΔCE agrees ($+0.178$ at $k=64$ vs. $\approx +0.14$ for the SAE). Low FVU therefore cannot validate provenance, and we score attribution directly against the ledger.

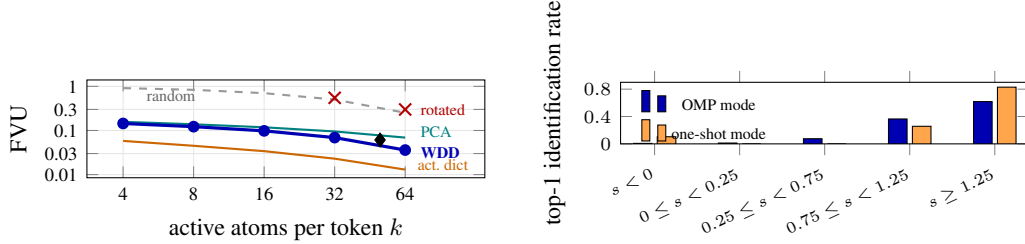


Figure 1: **Left:** reconstruction of GPT-2 layer-6 states. The training-free weight dictionary tracks a trained SAE’s FVU (diamond: a literature value from a different corpus [15]), and one Gram-preserving rotation of all atoms collapses it to the random floor (crosses). **Right:** identification rate of dominant writes by survival s . Cancelled directions ($s < 0.25$) are found near 1% of the time vs. 60–83% for reinforced ones.

Table 1: Four further families (middle layer). The rotation gap and the estimation/identification split replicate. Qwen2.5’s aggregate reflects heavy-tail sensitivity (see text). Recall: OMP-mode top-1.

Model	FVU@32/@64	rotated @32/@64	r on identified	top-1 recall
GPT-2 small (124M)	0.069/0.036	0.549/0.299	0.998	0.647
SmolLM2-135M	0.006/0.002	0.424/0.184	1.000	0.673
Qwen2.5-0.5B	0.018/0.010	0.544/0.312	0.798	0.482
Pythia-410m	0.051/0.032	0.606/0.383	0.987	0.488
OLMo-1B	0.018/0.014	0.781/0.621	0.997	0.713

4.2 Attribution and the Observability Ceiling

A token’s largest true write is recovered in OMP’s $k=64$ support 64.7% of the time. One-shot ranking does better (76.6%): OMP’s orthogonalization explains away correlated true writers once the state is nearly reconstructed. An identified dominant write’s coefficient is recovered at $r = 0.998$. Counting misses as zeros, the correlation is 0.648. In GPT-2 both figures scale with write magnitude (within-decile correlation below 0.28 outside the top decile). Gated architectures escape this law (Sec. 4.3).

Identification is limited neither by atom duplication (clustering near-duplicates moves recall $0.647 \rightarrow 0.664$) nor write age (AUC 0.51), but by observability (Fig. 1, right). Given the ledger coefficient, WDD certifies when a direction has been cancelled. Two oracle baselines locate the failure. On synthetic states built purely from each token’s MLP writes, recall is essentially unchanged (0.62 OMP, 0.67 one-shot). The ceiling belongs to this measurement geometry, with subsequent native writes cancelling a component along its own direction and defeating projection-based recovery. Conversely, refitting on the true top-3 atoms alone yields $r = 0.55$ (0.13 on erased writes). Joint sparse estimation over the full support is what delivers $r = 0.998$. Cancellation is directional, and information often survives it. A leakage-controlled probe on the full state recovers erased writers’ activations at median out-of-sample R^2 0.40 (intact: 0.59), and injected atoms are re-recovered at slope 0.90.

4.3 Generality and Blind Calibration

Decile stratification of Table 1 reveals an architectural divide: the gated-MLP families hold within-decile correlations of 0.89–1.00 in all magnitude deciles (unconditional r up to 0.97), while GPT-2 and Pythia, both non-gated, are magnitude-carried. Massive channels dominate raw FVU (the top-5 coordinates carry 91–98% of variance here), yet the advantage is not channel-bound. Off those coordinates the weight dictionary reaches FVU 0.30–0.54 vs. 3.0–13.5 rotated. The evaluation replicates on Pile (FVU@32 0.080 vs. 0.518 rotated; recall 0.562). Both of OLMo-1B’s published super weights [17] (down-projection [1764, 1710] and [1764, 8041], layers 1 and 2) emerge blind, each as the highest-norm atom of its block, with 97.9% and 92.7% of its energy in channel 1764.

4.4 Write Roles and Null Controls

The ledger played a held-out oracle’s role above, while WDD’s inference used only checkpoint and state. Discovery below works deliberately instrumented, reading recorded writes along the dictionary’s directions, with inversion mode the same instrument without hooks. On WikiText the taxonomy singles out GPT-2 small’s block 3: the only negative median birth-survival (-0.09) and the largest counter-aligned fraction (53%) in all pooled magnitude quintiles. Concentration analysis relocates the unit of counter-writing to specific neurons.

Random replacement directions collapse everything. Shuffled write–state pairings (marginals preserved) show that chance alone yields counter fractions of 51–72% (centered states anti-align with frequent writers), and each block’s real fraction sits below its own null, so we discard raw counter fraction. Anchoring separates real writes from both nulls: 24% of block-3 writes are born at their own survival anchor (nulls: 1.6% and 5.7%), and reinforcing mass reaches 61–75% in amplifier blocks (nulls: at most 6%). Within block 3, the erased subset is born at survival -0.63 (the block median pools it with intact writes), opposing the stream from the first moment rather than being consumed later. The profile is corpus-dependent (on Pile, block 3’s median turns positive): roles describe model-on-distribution.

4.5 The Channel-447 Circuit

Block 3’s dominant counter-writer is neuron #614, with 88% of its write energy in channel 447, GPT-2’s massive activation channel [16] with outlier-scale activations. Its atom unembeds to function words (8 of 8 top tokens), yet correlation calls it open-loop ($r=0.011$ along its own axis). Interventions test that verdict.

Dose-response. Scaling channel 447 at the block-2/3 boundary produces a block-wide inverse response (median active neuron $+9\%/-12\%$ at $0.5 \times /1.5 \times$), consistent with what LayerNorm renormalization predicts when an outlier coordinate is scaled. A frozen-normalization control separates the pathways: holding block 3’s pre-MLP LayerNorm statistics at clean-run values collapses the block-wide response (median $-0.5\%/-2\%$; inverse responders drop from 1,236 to 233) while #614 retains $+15.5\%/-12.3\%$ (94th percentile of the frozen distribution). About half its raw response ($+27.5\%/-22.1\%$, independently re-implemented) is renormalization, the rest content-driven. The upstream channel-builder’s response is zero to numerical precision, a placement check.

Group ablation and restoration. Ablating the four strongest channel-447 counter-writers (blocks 3–6) shifts the channel by the predicted -12.4 , yet only -1.49 remains at the layer-11 readout: 88% restored on average (a means-level result) by specific later neurons, not passive renormalization. Behavioral cost is strongly nonlinear in write energy: the largest counter-writer alone (65% of the set’s energy) costs $\Delta\text{CE } 0.004$, the top three 0.024 , saturating through all nine, while random 9-neuron sets cost under 0.001 . The function is collectively carried and redundant.

Disinhibition. The restoring neurons are channel-insensitive (gains near zero, wrong sign), yet their channel-447 writes rise 10–51% under group ablation. Write-editing locates the suppression: removing only the non-channel components of the group’s writes reproduces 75% of the reserve response; the channel component alone, 19%; #614’s component alone, none; the suppression is distributed across the group.

4.6 Nomination Comparisons

Do WDD nominations resolve into live circuits more often than those from standard screens? All arms share one battery, fixed in code before execution, including the step naming each nominee’s predicted partner. Studied and channel-loaded neurons are excluded everywhere. Tier 1 asks whether ablating a nominee shifts the state as its own recorded write predicts, and Tier 2 whether it responds when its predicted partner is ablated.

At $N=3$ per arm, WDD’s counter-energy screen went 0/3 on Tier 1 (one sign flip, traceable to the restoration effect above) and 3/3 on Tier 2, vs. 1/3 and 0/3 for a kurtosis-times-magnitude screen (Tier 2 alone: exact $p = 0.05$; both tiers: 0.20). Because that screen is marginal while the endpoint is relational, we reran at $N=10$ per arm with a relational baseline (maximum upstream activation correlation), and the early separation did not replicate. WDD passed Tier 2 at 8/10 and Tier 1 at 4/10, the outlier screen reached 6/10 and 2/10 (not significantly different), and the correlation baseline

163 matched WDD on Tier 2 (8/10) while passing Tier 1 at 9/10. Coupled pairs are commonplace, and
164 a one-line correlation screen finds them as reliably as WDD. The write-direction screen instead
165 adds mechanism-level content: a named opposed write, a predicted partner, and a role hypothesis
166 interventions can test.

167 5 Discussion and Limitations

168 The ceiling is identification (recall 0.48–0.71), driven by write-population self-cancellation, and
169 non-gated coefficients are reliable only in proportion to write magnitude. The information-survival
170 probe fails for rare firers out of distribution, and role statistics depend on frame and corpus. The
171 circuit study covers one model and largely one channel, the frozen-normalization control spans only
172 block 3’s MLP path, and restoration is means-level. The comparisons are small, their endpoint
173 favors relational screens, and Tier 1 penalizes the circuits Tier 2 rewards. All results come from one
174 implementation, bounded by matched controls, a line-by-line audit, and re-implementations of four
175 headline numbers, with code and results released.

176 6 Related Work

177 **Write persistence.** Janiak et al. [10] define the projection ratio, find anti-aligned cleanup heads, and
178 intervene on writer–cleaner pairs. We extend erasure to role characterization. **The model’s own**
179 **vectors.** Geva et al. [8] read MLP value vectors by vocabulary projection, Dar et al. [6] interpret
180 weights in embedding space, and Ye et al. [12] find such vectors rival SAE features. Direct logit
181 attribution [7] requires the full forward pass. **Learned dictionaries.** SAEs [3, 5] learn atoms from
182 activations, and ITDA [9] harvests its dictionary from them, a complementary activation-space
183 analogue. **Weight-space interpretability.** ROTATE [13], the closest neighbor, decomposes weight
184 vectors themselves, where WDD decomposes observed states. SITH [14] decomposes CLIP attention
185 in concept space, parameter decomposition splits the parameters [2, 4], and interference-weight
186 analysis [1] warns weight-implied links need not be functional. Massive activations and super weights
187 are calibration targets [16, 11, 17].

188 7 Conclusion

189 We introduced weight-dictionary decomposition, which reads residual states over a provenance-
190 bearing dictionary built from the model’s own weights, validated it across five families, and resolved
191 GPT-2’s apparent channel eraser into a maintenance participant. Geometry nominates, and interven-
192 tion assigns meaning.

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223 Appendix (assembled from released artifacts)

224 **A:** dictionary construction per architecture, with the exact atom census (50,257 token + 1,024
225 position + 7 blocks $\times$ (3,072 MLP + 768 attention-SVD + 2 bias) = 78,175; overcomplete means
226 $m/d \approx 102$). **B:** full role maps, null tables, quintile analyses, cross-domain replication. **C:** full
227 pre-registration text and per-nomination results for both comparisons. **D:** population dose-response
228 distributions (frozen and unfrozen), write-editing conditions, restoration roster. **E:** oracle baselines
229 and dictionary coherence; refuted hypotheses; self-cancellation ledger. **F:** audit and re-implementation
230 reports; probe fragility.